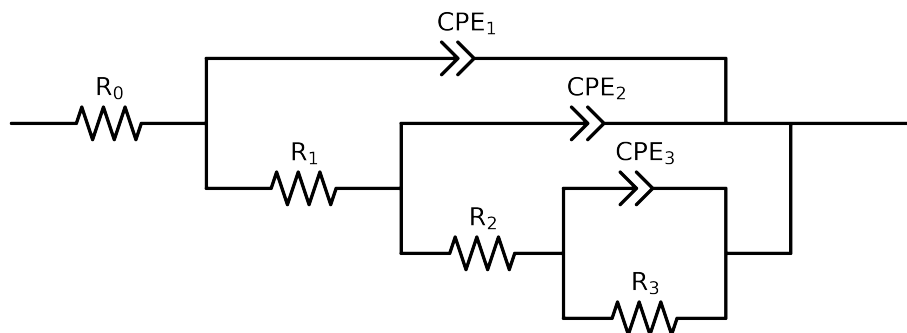


Comparative EIS Kinetics Report

Model Structure: R0-p(R1-p(R2-p(R3,CPE3),CPE2),CPE1)

Used data: original data



Element	Unit	Bounds	Cauvel_ PA_ 168H	Cauvel_ PA_ 24H	Cauvel_ PA_ 72H
R_0	Ω	$[1.0e+00, 1.0e+03]$	$6.85e+01 \pm 7.3e-01$	$6.85e+01 \pm 5.3e-01$	$6.26e+01 \pm 7.7e-01$
R_1	Ω	$[1.0e+00, 1.0e+08]$	$3.79e+01 \pm 1.0e+01$	$1.80e+04 \pm 1.9e+03$	$1.38e+02 \pm 6.0e+01$
R_2	Ω	$[1.0e+00, 1.0e+08]$	$6.08e+03 \pm 2.4e+02$	$1.00e+04 \pm 1.0e+04$	$1.34e+04 \pm 7.4e+02$
R_3	Ω	$[1.0e+00, 1.0e+08]$	$1.52e+04 \pm 7.2e+02$	$7.43e+03 \pm 9.8e+03$	$2.88e+04 \pm 2.5e+03$
Q_3	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-02]$	$2.06e-04 \pm 9.7e-06$	$2.17e-04 \pm 4.7e-04$	$1.21e-04 \pm 1.1e-05$
n_3	-	$[5.0e-01, 1.0e+00]$	$6.29e-01 \pm 2.1e-02$	$7.66e-01 \pm 3.1e-01$	$6.27e-01 \pm 3.8e-02$
Q_2	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-02]$	$1.76e-05 \pm 7.1e-06$	$2.78e-05 \pm 1.9e-05$	$9.14e-06 \pm 5.5e-06$
n_2	-	$[5.0e-01, 1.0e+00]$	$8.04e-01 \pm 3.3e-02$	$1.00e+00 \pm 3.9e-01$	$7.97e-01 \pm 5.0e-02$
Q_1	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$8.62e-06 \pm 8.0e-06$	$1.63e-05 \pm 4.7e-07$	$9.41e-06 \pm 6.1e-06$
n_1	-	$[5.0e-01, 1.0e+00]$	$7.84e-01 \pm 7.9e-02$	$7.47e-01 \pm 4.1e-03$	$7.85e-01 \pm 5.8e-02$
χ^2	-	-	$2.042e-04$	$7.180e-04$	$4.382e-04$

Analysis Results: Cauvel_PA_168H

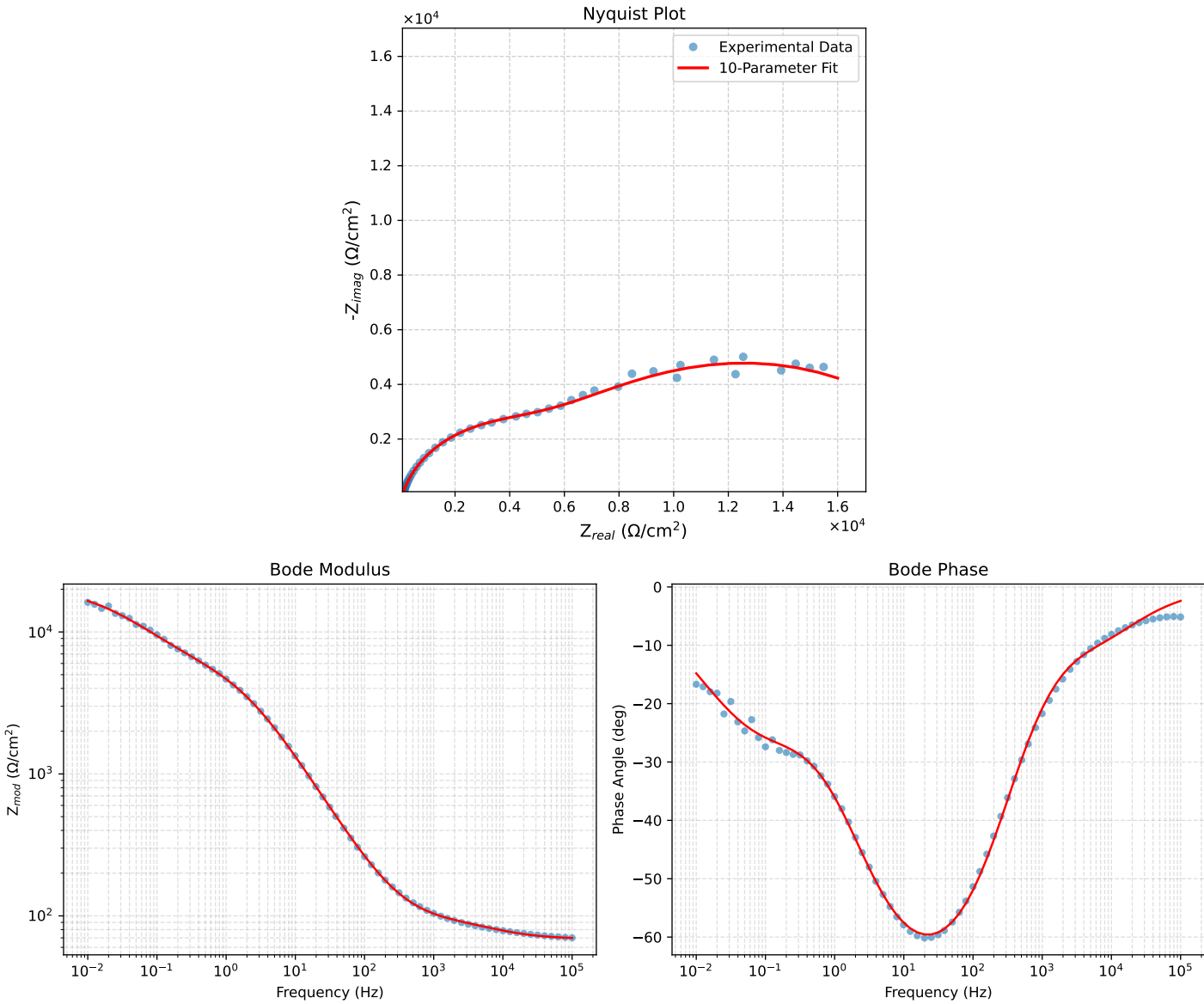


Figure 1: EIS Fit Results for Cauvel_PA_168H

Fitted Data: Cauvel_PA_168H

Frequency (Hz)	$Z_{real}(\Omega)$	$-Z_{imag}(\Omega)$	$Z_{mod}(\Omega)$	$Z_{phase}(\circ)$
1.0008e+05	6.9822e+01	2.9124e+00	6.9883e+01	-2.3886
7.9453e+04	7.0115e+01	3.4412e+00	7.0200e+01	-2.8098
6.3141e+04	7.0477e+01	4.0499e+00	7.0593e+01	-3.2888
5.0203e+04	7.0924e+01	4.7457e+00	7.1082e+01	-3.8281
3.9891e+04	7.1476e+01	5.5367e+00	7.1690e+01	-4.4295
3.1641e+04	7.2158e+01	6.4292e+00	7.2443e+01	-5.0915
2.5172e+04	7.2975e+01	7.4000e+00	7.3349e+01	-5.7902
2.0016e+04	7.3955e+01	8.4531e+00	7.4437e+01	-6.5206
1.5891e+04	7.5117e+01	9.5800e+00	7.5725e+01	-7.2679
1.2609e+04	7.6458e+01	1.0759e+01	7.7211e+01	-8.0103
1.0078e+04	7.7914e+01	1.1937e+01	7.8823e+01	-8.7103
8.0156e+03	7.9536e+01	1.3173e+01	8.0620e+01	-9.4043
6.3281e+03	8.1315e+01	1.4502e+01	8.2598e+01	-10.1120
5.0156e+03	8.3127e+01	1.5905e+01	8.4635e+01	-10.8315
3.9844e+03	8.4949e+01	1.7459e+01	8.6724e+01	-11.6141
3.1710e+03	8.6767e+01	1.9259e+01	8.8878e+01	-12.5145
2.5276e+03	8.8586e+01	2.1411e+01	9.1137e+01	-13.5878
1.9761e+03	9.0604e+01	2.4305e+01	9.3807e+01	-15.0165
1.5775e+03	9.2535e+01	2.7608e+01	9.6566e+01	-16.6125
1.2656e+03	9.4558e+01	3.1588e+01	9.9695e+01	-18.4722
9.9826e+02	9.6963e+01	3.6882e+01	1.0374e+02	-20.8252
7.9688e+02	9.9548e+01	4.3066e+01	1.0846e+02	-23.3940
6.2779e+02	1.0271e+02	5.1082e+01	1.1471e+02	-26.4423
5.0551e+02	1.0608e+02	5.9918e+01	1.2183e+02	-29.4590
3.9800e+02	1.1049e+02	7.1719e+01	1.3173e+02	-32.9864
3.1550e+02	1.1565e+02	8.5620e+01	1.4390e+02	-36.5134
2.5240e+02	1.2162e+02	1.0167e+02	1.5851e+02	-39.8942
1.9862e+02	1.2940e+02	1.2239e+02	1.7811e+02	-43.4063
1.5836e+02	1.3837e+02	1.4592e+02	2.0110e+02	-46.5211
1.2556e+02	1.4962e+02	1.7475e+02	2.3005e+02	-49.4306
1.0045e+02	1.6285e+02	2.0776e+02	2.6398e+02	-51.9097
7.9003e+01	1.8041e+02	2.5012e+02	3.0840e+02	-54.1971
6.3345e+01	2.0039e+02	2.9642e+02	3.5780e+02	-55.9401
5.0223e+01	2.2633e+02	3.5388e+02	4.2007e+02	-57.3977
3.8422e+01	2.6433e+02	4.3323e+02	5.0750e+02	-58.6113
3.1250e+01	3.0112e+02	5.0540e+02	5.8830e+02	-59.2132
2.4934e+01	3.5075e+02	5.9658e+02	6.9205e+02	-59.5470
1.9862e+01	4.1314e+02	7.0272e+02	8.1517e+02	-59.5482
1.5625e+01	4.9587e+02	8.3155e+02	9.6818e+02	-59.1916
1.2401e+01	5.9593e+02	9.7268e+02	1.1407e+03	-58.5053
9.9311e+00	7.1510e+02	1.1239e+03	1.3321e+03	-57.5335
7.9449e+00	8.6188e+02	1.2902e+03	1.5516e+03	-56.2558
6.3174e+00	1.0454e+03	1.4732e+03	1.8064e+03	-54.6414
5.0080e+00	1.2691e+03	1.6670e+03	2.0951e+03	-52.7163
3.9457e+00	1.5412e+03	1.8680e+03	2.4217e+03	-50.4756
3.1587e+00	1.8339e+03	2.0506e+03	2.7510e+03	-48.1925
2.5040e+00	2.1768e+03	2.2289e+03	3.1155e+03	-45.6769
1.9981e+00	2.5410e+03	2.3844e+03	3.4845e+03	-43.1789
1.5847e+00	2.9388e+03	2.5227e+03	3.8731e+03	-40.6433
1.2669e+00	3.3375e+03	2.6360e+03	4.2529e+03	-38.3021
9.9904e-01	3.7681e+03	2.7384e+03	4.6581e+03	-36.0074
7.9234e-01	4.1902e+03	2.8276e+03	5.0550e+03	-34.0123
6.3345e-01	4.5972e+03	2.9113e+03	5.4415e+03	-32.3455
5.0403e-01	5.0132e+03	3.0021e+03	5.8434e+03	-30.9152
4.0064e-01	5.4353e+03	3.1055e+03	6.2599e+03	-29.7418
3.1672e-01	5.8779e+03	3.2290e+03	6.7064e+03	-28.7818
2.5202e-01	6.3257e+03	3.3688e+03	7.1668e+03	-28.0377
2.0032e-01	6.8004e+03	3.5287e+03	7.6614e+03	-27.4247
1.5890e-01	7.3124e+03	3.7074e+03	8.1985e+03	-26.8852
1.2601e-01	7.8649e+03	3.8985e+03	8.7781e+03	-26.3670
1.0016e-01	8.4565e+03	4.0924e+03	9.3947e+03	-25.8237
7.9449e-02	9.1014e+03	4.2831e+03	1.0059e+04	-25.2013
6.3174e-02	9.7873e+03	4.4562e+03	1.0754e+04	-24.4798
5.0134e-02	1.0524e+04	4.6033e+03	1.1487e+04	-23.6256
3.9826e-02	1.1295e+04	4.7113e+03	1.2238e+04	-22.6419
3.1630e-02	1.2094e+04	4.7712e+03	1.3001e+04	-21.5291
2.5121e-02	1.2908e+04	4.7757e+03	1.3763e+04	-20.3038
1.9955e-02	1.3720e+04	4.7216e+03	1.4509e+04	-18.9908
1.5852e-02	1.4515e+04	4.6094e+03	1.5229e+04	-17.6176
1.2591e-02	1.5280e+04	4.4433e+03	1.5913e+04	-16.2139
1.0001e-02	1.6001e+04	4.2311e+03	1.6551e+04	-14.8112

Analysis Results: Cauvel_PA_24H_

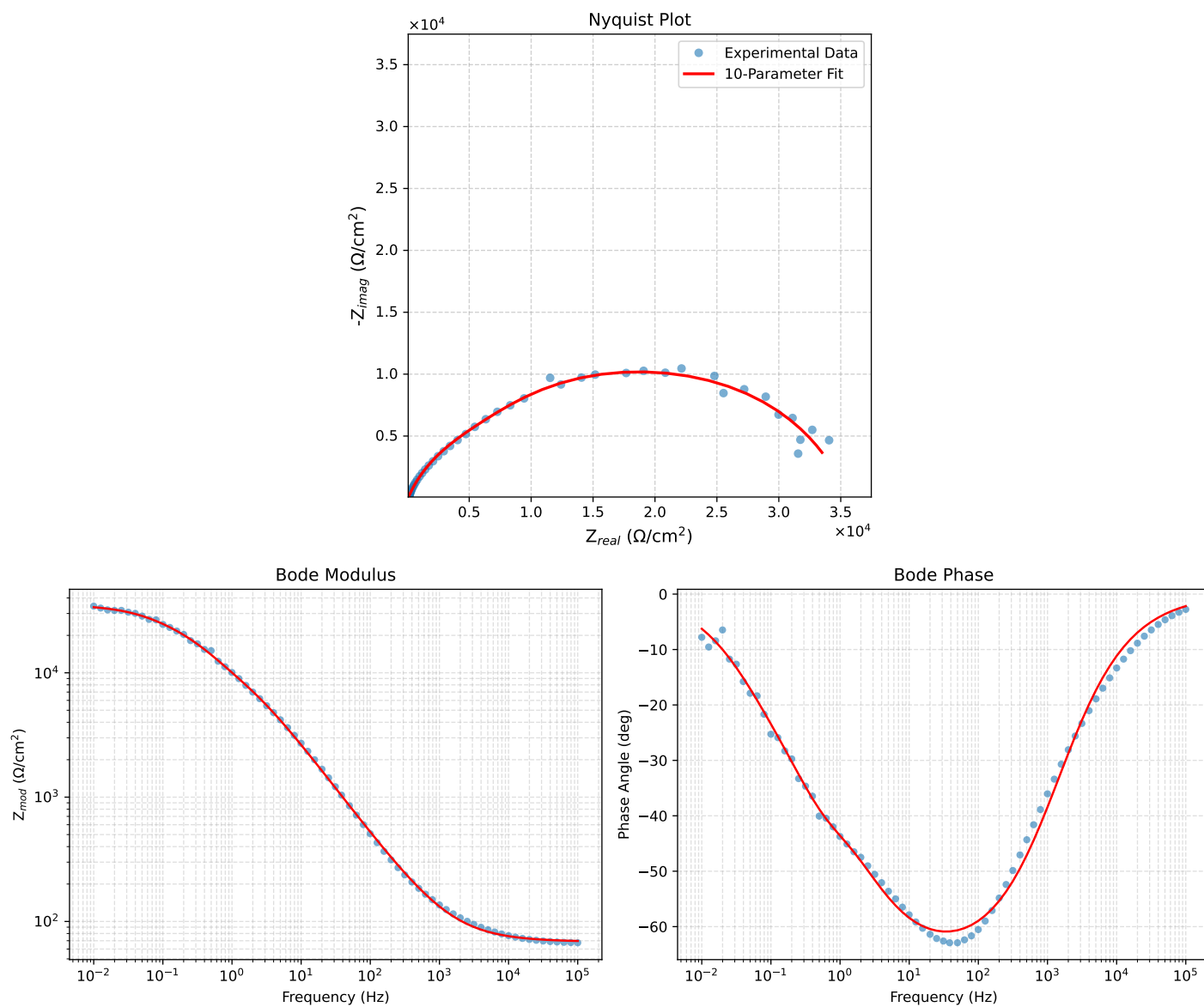


Figure 2: EIS Fit Results for Cauvel_PA_24H_

Fitted Data: Cauvel_PA_24H

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^{\circ}$)
1.0002e+05	6.9633e+01	2.6400e+00	6.9683e+01	-2.1712
7.9512e+04	6.9840e+01	3.1335e+00	6.9911e+01	-2.5690
6.3105e+04	7.0088e+01	3.7239e+00	7.0187e+01	-3.0414
5.0215e+04	7.0380e+01	4.4169e+00	7.0518e+01	-3.5911
3.9902e+04	7.0727e+01	5.2442e+00	7.0921e+01	-4.2405
3.1699e+04	7.1141e+01	6.2276e+00	7.1413e+01	-5.0029
2.5137e+04	7.1636e+01	7.4055e+00	7.2018e+01	-5.9021
1.9980e+04	7.2219e+01	8.7903e+00	7.2752e+01	-6.9398
1.5879e+04	7.2911e+01	1.0435e+01	7.3654e+01	-8.1452
1.2598e+04	7.3741e+01	1.2404e+01	7.4777e+01	-9.5486
1.0020e+04	7.4715e+01	1.4717e+01	7.6151e+01	-11.1429
7.9282e+03	7.5901e+01	1.7527e+01	7.7898e+01	-13.0027
6.3079e+03	7.7278e+01	2.0788e+01	8.0025e+01	-15.0564
5.0347e+03	7.8887e+01	2.4597e+01	8.2633e+01	-17.3171
3.9931e+03	8.0853e+01	2.9240e+01	8.5978e+01	-19.8825
3.1441e+03	8.3272e+01	3.4948e+01	9.0308e+01	-22.7668
2.5195e+03	8.5937e+01	4.1222e+01	9.5312e+01	-25.6261
2.0008e+03	8.9227e+01	4.8952e+01	1.0177e+02	-28.7501
1.5807e+03	9.3237e+01	5.8349e+01	1.0999e+02	-32.0390
1.2536e+03	9.7944e+01	6.9345e+01	1.2001e+02	-35.2986
1.0007e+03	1.0339e+02	8.2009e+01	1.3196e+02	-38.4226
7.9003e+02	1.1019e+02	9.7774e+01	1.4731e+02	-41.5838
6.2881e+02	1.1803e+02	1.1585e+02	1.6539e+02	-44.4661
5.0223e+02	1.2721e+02	1.3689e+02	1.8688e+02	-47.0992
4.0015e+02	1.3825e+02	1.6202e+02	2.1299e+02	-49.5254
3.1550e+02	1.5208e+02	1.9321e+02	2.4588e+02	-51.7929
2.5202e+02	1.6771e+02	2.2812e+02	2.8314e+02	-53.6781
2.0032e+02	1.8678e+02	2.7024e+02	3.2851e+02	-55.3492
1.5801e+02	2.1047e+02	3.2182e+02	3.8454e+02	-56.8158
1.2556e+02	2.3808e+02	3.8098e+02	4.4925e+02	-57.9978
9.9734e+01	2.7133e+02	4.5087e+02	5.2621e+02	-58.9609
7.9449e+01	3.1082e+02	5.3209e+02	6.1622e+02	-59.7087
6.2919e+01	3.5971e+02	6.3010e+02	7.2554e+02	-60.2787
4.9867e+01	4.1875e+02	7.4494e+02	8.5457e+02	-60.6584
3.8422e+01	5.0012e+02	8.9742e+02	1.0274e+03	-60.8700
3.1250e+01	5.7857e+02	1.0387e+03	1.1889e+03	-60.8807
2.4934e+01	6.8164e+02	1.2164e+03	1.3944e+03	-60.7351
2.0032e+01	8.0250e+02	1.4147e+03	1.6265e+03	-60.4360
1.5625e+01	9.7028e+02	1.6741e+03	1.9350e+03	-59.9045
1.2467e+01	1.1571e+03	1.9442e+03	2.2625e+03	-59.2396
9.9734e+00	1.3806e+03	2.2449e+03	2.6355e+03	-58.4082
7.9719e+00	1.6510e+03	2.5813e+03	3.0642e+03	-57.3968
6.3516e+00	1.9802e+03	2.9571e+03	3.5589e+03	-56.1924
5.0134e+00	2.3898e+03	3.3825e+03	4.1415e+03	-54.7585
3.9860e+00	2.8582e+03	3.8232e+03	4.7735e+03	-53.2184
3.1758e+00	3.3954e+03	4.2819e+03	5.4647e+03	-51.5865
2.5161e+00	4.0218e+03	4.7698e+03	6.2391e+03	-49.8630
1.9955e+00	4.7199e+03	5.2723e+03	7.0763e+03	-48.1640
1.5879e+00	5.4801e+03	5.7877e+03	7.9705e+03	-46.5637
1.2581e+00	6.3341e+03	6.3416e+03	8.9631e+03	-45.0337
9.9777e-01	7.2794e+03	6.9275e+03	1.0049e+04	-43.5810
7.9274e-01	8.3349e+03	7.5384e+03	1.1238e+04	-42.1273
6.2903e-01	9.5378e+03	8.1612e+03	1.2553e+04	-40.5526
4.9888e-01	1.0897e+04	8.7577e+03	1.3980e+04	-38.7876
3.9860e-01	1.2354e+04	9.2695e+03	1.5445e+04	-36.8818
3.1672e-01	1.3966e+04	9.6921e+03	1.7000e+04	-34.7597
2.5148e-01	1.5670e+04	9.9901e+03	1.8584e+04	-32.5185
1.9998e-01	1.7413e+04	1.0152e+04	2.0156e+04	-30.2439
1.5879e-01	1.9185e+04	1.0181e+04	2.1718e+04	-27.9536
1.2601e-01	2.0953e+04	1.0076e+04	2.3250e+04	-25.6837
1.0016e-01	2.2675e+04	9.8444e+03	2.4720e+04	-23.4684
7.9503e-02	2.4348e+04	9.4850e+03	2.6130e+04	-21.2836
6.3140e-02	2.5930e+04	9.0077e+03	2.7450e+04	-19.1563
5.0123e-02	2.7399e+04	8.4253e+03	2.8665e+04	-17.0931
3.9833e-02	2.8720e+04	7.7650e+03	2.9751e+04	-15.1294
3.1638e-02	2.9887e+04	7.0517e+03	3.0708e+04	-13.2758
2.5126e-02	3.0893e+04	6.3178e+03	3.1533e+04	-11.5579
1.9957e-02	3.1742e+04	5.5931e+03	3.2231e+04	-9.9934
1.5849e-02	3.2446e+04	4.9002e+03	3.2814e+04	-8.5883
1.2590e-02	3.3022e+04	4.2567e+03	3.3296e+04	-7.3451
1.0001e-02	3.3491e+04	3.6712e+03	3.3691e+04	-6.2557

Analysis Results: Cauvel_PA_72H_

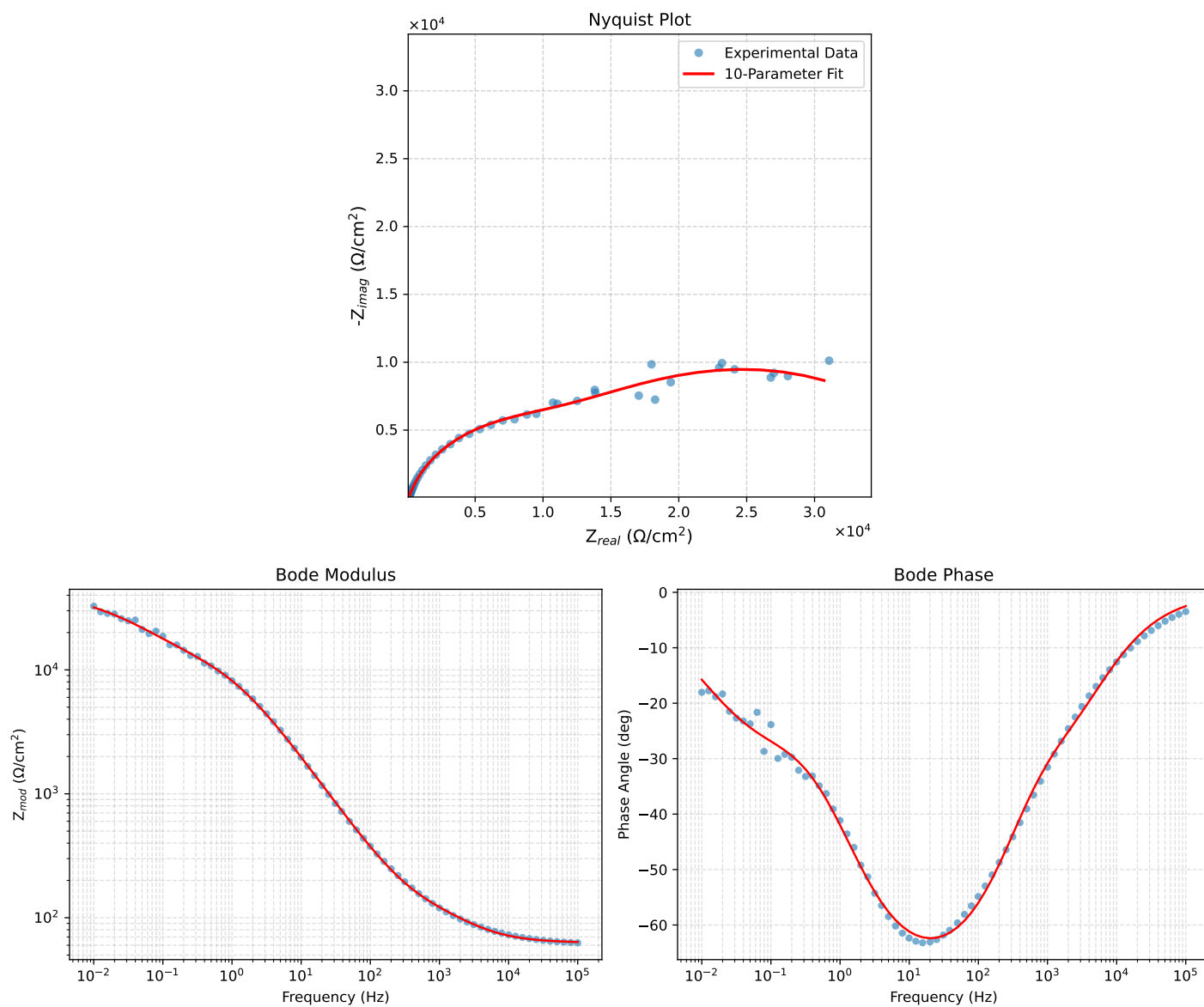


Figure 3: EIS Fit Results for Cauvel_PA_72H_

Fitted Data: Cauvel_PA_72H

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^\circ$)
1.0002e+05	6.3648e+01	2.7677e+00	6.3708e+01	-2.4899
7.9512e+04	6.3862e+01	3.3039e+00	6.3948e+01	-2.9615
6.3105e+04	6.4126e+01	3.9464e+00	6.4247e+01	-3.5216
5.0215e+04	6.4444e+01	4.7006e+00	6.4615e+01	-4.1718
3.9902e+04	6.4835e+01	5.5993e+00	6.5077e+01	-4.9359
3.1699e+04	6.5316e+01	6.6629e+00	6.5655e+01	-5.8247
2.5137e+04	6.5911e+01	7.9274e+00	6.6386e+01	-6.8583
1.9980e+04	6.6635e+01	9.3975e+00	6.7295e+01	-8.0274
1.5879e+04	6.7527e+01	1.1116e+01	6.8436e+01	-9.3482
1.2598e+04	6.8633e+01	1.3129e+01	6.9877e+01	-10.8298
1.0020e+04	6.9974e+01	1.5428e+01	7.1654e+01	-12.4335
7.9282e+03	7.1649e+01	1.8120e+01	7.3904e+01	-14.1928
6.3079e+03	7.3631e+01	2.1106e+01	7.6596e+01	-15.9950
5.0347e+03	7.5965e+01	2.4412e+01	7.9791e+01	-17.8153
3.9931e+03	7.8796e+01	2.8200e+01	8.3690e+01	-19.6917
3.1441e+03	8.2191e+01	3.2540e+01	8.8398e+01	-21.5992
2.5195e+03	8.5765e+01	3.6990e+01	9.3402e+01	-23.3306
2.0008e+03	8.9899e+01	4.2134e+01	9.9282e+01	-25.1116
1.5807e+03	9.4519e+01	4.8060e+01	1.0604e+02	-26.9520
1.2536e+03	9.9417e+01	5.4749e+01	1.1350e+02	-28.8418
1.0007e+03	1.0450e+02	6.2331e+01	1.2168e+02	-30.8141
2.5202e+02	1.4813e+02	1.5533e+02	2.1464e+02	-46.3605
2.0032e+02	1.5957e+02	1.8398e+02	2.4354e+02	-49.0630
1.5801e+02	1.7369e+02	2.1975e+02	2.8011e+02	-51.6772
1.2556e+02	1.9015e+02	2.6151e+02	3.2333e+02	-53.9783
9.9734e+01	2.1007e+02	3.1166e+02	3.7585e+02	-56.0182
7.9449e+01	2.3395e+02	3.7082e+02	4.3845e+02	-57.7529
6.2919e+01	2.6385e+02	4.4326e+02	5.1584e+02	-59.2374
4.9867e+01	3.0046e+02	5.2938e+02	6.0870e+02	-60.4225
3.8422e+01	3.5175e+02	6.4546e+02	7.3509e+02	-61.4112
3.1250e+01	4.0207e+02	7.5452e+02	8.5497e+02	-61.9476
2.4934e+01	4.6935e+02	8.9365e+02	1.0094e+03	-62.2916
2.0032e+01	5.4975e+02	1.0511e+03	1.1862e+03	-62.3889
1.5625e+01	6.6384e+02	1.2602e+03	1.4244e+03	-62.2212
1.2467e+01	7.9395e+02	1.4815e+03	1.6808e+03	-61.8125
9.9734e+00	9.5340e+02	1.7319e+03	1.9770e+03	-61.1671
7.9719e+00	1.1514e+03	2.0166e+03	2.3222e+03	-60.2743
6.3516e+00	1.3996e+03	2.3400e+03	2.7266e+03	-59.1151
5.0134e+00	1.7186e+03	2.7119e+03	3.2106e+03	-57.6359
3.9860e+00	2.0969e+03	3.1021e+03	3.7443e+03	-55.9422
3.1758e+00	2.5476e+03	3.5098e+03	4.3369e+03	-54.0251
2.5161e+00	3.0945e+03	3.9380e+03	5.0084e+03	-51.8394
1.9955e+00	3.7267e+03	4.3607e+03	5.7362e+03	-49.4820
1.5879e+00	4.4325e+03	4.7591e+03	6.5035e+03	-47.0352
1.2581e+00	5.2267e+03	5.1834e+03	7.3260e+03	-44.4841
9.9777e-01	6.0782e+03	5.4656e+03	8.1742e+03	-41.9626
7.9274e-01	6.9663e+03	5.7534e+03	9.0350e+03	-39.5531
6.2903e-01	7.8868e+03	6.0058e+03	9.9132e+03	-37.2894
4.9888e-01	8.8248e+03	6.2327e+03	1.0804e+04	-35.2327
3.9860e-01	9.7418e+03	6.4412e+03	1.1679e+04	-33.4724
3.1672e-01	1.0691e+04	6.6577e+03	1.2594e+04	-31.9127
2.5148e-01	1.1659e+04	6.8006e+03	1.3543e+04	-30.5834
1.9998e-01	1.2647e+04	7.1459e+03	1.4526e+04	-29.4674
1.5879e-01	1.3682e+04	7.4307e+03	1.5569e+04	-28.5071
1.2601e-01	1.4773e+04	7.7422e+03	1.6679e+04	-27.6577
1.0016e-01	1.5924e+04	8.0695e+03	1.7852e+04	-26.8741
7.9503e-02	1.7159e+04	8.4040e+03	1.9106e+04	-26.0947
6.3140e-02	1.8475e+04	8.7257e+03	2.0432e+04	-25.2816
5.0123e-02	1.9877e+04	9.0153e+03	2.1826e+04	-24.3967
3.9833e-02	2.1350e+04	9.2492e+03	2.3268e+04	-23.4230
3.1638e-02	2.2891e+04	9.4084e+03	2.4749e+04	-22.3434
2.5126e-02	2.4476e+04	9.4741e+03	2.6245e+04	-21.1607
1.9957e-02	2.6077e+04	9.4340e+03	2.7731e+04	-19.8885
1.5849e-02	2.7669e+04	9.2823e+03	2.9185e+04	-18.5454
1.2590e-02	2.9218e+04	9.0219e+03	3.0580e+04	-17.1594
1.0001e-02	3.0700e+04	8.6620e+03	3.1899e+04	-15.7563